

KiCad for Conductivity Cell Design

José Díaz-Milhet, PhD
Postdoc at MWET (UT Austin) and LBNL-ALS

October 2, 2026

Printed circuit boards (PCBs) make excellent, inexpensive and highly reproducible electrode substrates for electrochemical measurements: the electrode geometry is defined by lithography to within tens of micrometers, the exposed pads can be gold-plated, and a batch of identical cells costs a few dollars. This guide shows how to download and install KiCad, a free and open-source PCB design suite, explains the handful of concepts needed to draw a simple electrode board, and walks through two worked examples: a through-plane and an in-plane cell for measuring the ionic conductivity of membranes.

Contents

1	Why Use a PCB as an Electrochemical Cell?	2
2	Downloading and Installing KiCad	2
3	PCB Primer	2
3.1	The project manager and its tools	2
3.2	Layers	2
3.3	Units and grid	3
3.4	Exact dimensions come from the Properties dialog	3
3.5	Pads belong to footprints	3
4	Step by Step Small Workflow	4
4.1	Do I need a schematic?	4
4.2	Steps	4
5	Example 1: Through-Plane Conductivity Cell	5
6	Example 2: In-Plane Conductivity Cell	6
7	Ordering Boards	7
7.1	What every manufacturer needs	7
7.2	Choices that matter for electrochemistry	7
7.3	Example: OSH Park	8
7.4	Other manufacturers	8
8	Further Reading	8

1 Why Use a PCB as an Electrochemical Cell?

A conductivity cell needs electrodes of accurately known geometry, a rigid flat support and reliable connections to the potentiostat. A PCB provides all three:

- **Defined geometry.** Electrode area and spacing—the quantities that enter the cell constant—are set by the manufacturer’s photolithography, not by hand.
- **Inert surface.** With an ENIG (electro-less nickel immersion gold) finish, all exposed copper is covered by nickel and a thin layer of gold.
- **Built-in insulation.** The solder mask covers the traces, so only the electrode and the contact pad touch the sample or the clips.
- **Low cost and reproducibility.** Boards are ordered in batches, so every lab member, or every collaborating lab, can use an identical cell.

KiCad is well suited to this job because it is free, runs on Windows, macOS and Linux, and its native files are accepted directly by several board manufacturers.

2 Downloading and Installing KiCad

KiCad is distributed from its official site, <https://www.kicad.org/download/>. Always download from there. This guide was written for KiCad 10 (version 10.0.6, the stable release in October 2026); versions 8 and 9 look and behave almost identically.

First launch. KiCad asks whether to import settings from an earlier version or start with defaults (choose defaults on a new machine), and how to set up the global symbol and footprint library tables. Choose the recommended option, which copies the default tables, so that the standard libraries (including the mounting-hole footprints used below) are available.

3 PCB Primer

3.1 The project manager and its tools

Opening KiCad shows the *project manager*. A project is a folder containing a project file (`.kicad_pro`), a schematic (`.kicad_sch`) and a board layout (`.kicad_pcb`). The tools that matter here are:

Tool	What it is for
PCB Editor	Drawing the physical board: outline, copper, mask openings, holes. This is where almost all the work for an electrode board happens.
Footprint Editor	Creating reusable pads and holes (e.g. a disc electrode or a slot) that can be placed on any board.
Schematic Editor	Drawing the electrical circuit. Optional for a simple electrode board (Section 4.1).
Gerber Viewer	Checking the exported manufacturing files before ordering.

3.2 Layers

A board is described as a stack of layers, selected in the *Appearance* panel on the right of the PCB Editor. Whatever you draw goes on the *active* layer (marked by a small arrow in the list). Only a few layers matter for an electrode board (Table 1 and Fig. 1).

Table 1: The layers used for an electrode board.

Layer	Meaning
F.Cu, B.Cu	Copper on the front and back. Pads, electrodes and tracks.
F.Mask, B.Mask	Solder mask. Negative: a shape drawn here is an <i>opening</i> in the mask, i.e. exposed (gold-plated) metal.
Edge.Cuts	Board outline and internal cut-outs. Must be closed shapes.
F.Silkscreen, B.Silkscreen	Printed labels (electrode numbers, version, date).
User.Drawings, User.Comments	Notes and dimensions for yourself; not manufactured.

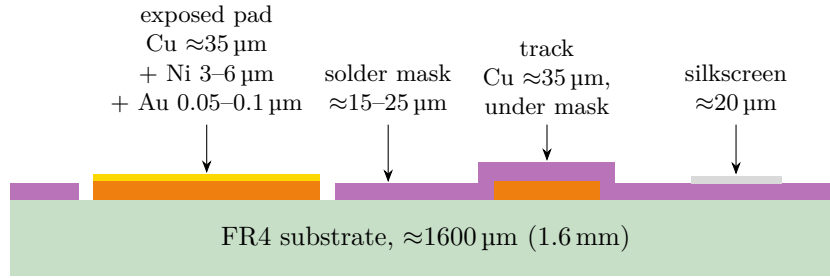


Figure 1: Cross-section of a board (not to scale) with typical layer thicknesses for a standard 1 oz copper, ENIG-finished board. The solder mask covers everything except where a shape has been drawn on the mask layer; the surface finish is applied only to exposed copper. Exact values vary by manufacturer.

3.3 Units and grid

Work in mm: toggle units with the **mm** button on the left toolbar or **Ctrl+U**. Choose a grid from the drop-down list in the top toolbar (e.g. 0.5 mm); everything you draw snaps to it, which makes symmetric layouts easy. The status bar at the bottom shows the absolute cursor position and the distance (**dx**, **dy**, **dist**) from a local origin that you reset by pressing **Space**. This is the quickest way to measure electrode spacing.

3.4 Exact dimensions come from the Properties dialog

Do not try to draw to exact size with the mouse. Draw the shape approximately, select it, press **E** and type the exact coordinates, sizes and line widths.

3.5 Pads belong to footprints

In KiCad a *pad* (a piece of copper with a matching mask opening, optionally with a hole) lives inside a *footprint*. For an electrode board there are two ways to make an electrode:

1. **Footprint route (recommended for reuse).** In the Footprint Editor, create a new footprint containing one surface-mount pad of the required shape and size, enabled on **F.Cu** and **F.Mask** only (untick **F.Paste**). Place it on the board with **A**.
2. **Graphic route (quickest).** In the PCB Editor, draw a filled circle or rectangle on **F.Cu**, then an identical shape on **F.Mask** to expose it.

4 Step by Step Small Workflow

4.1 Do I need a schematic?

Short answer, NO! You can just open the PCB Editor and draw directly. The price is that KiCad does not know which pads are meant to be connected, so you must check connectivity by eye. For anything with components (connectors, resistors, sensors), draw a schematic first and use Tools ► Update PCB from Schematic.

4.2 Steps

1. **Create the project.** In the project manager, File ► New Project. Open the .kicad_pcb file to start the PCB Editor.
2. **Set the manufacturer's limits.** File ► Board Setup ► Design Rules ► Constraints. Enter the minimum track width, clearance and hole size of the manufacturer you intend to use (Section 7), so that the design-rule check is meaningful.
3. **Draw the board outline.** Make `Edge.Cuts` the active layer and draw a rectangle with the rectangle tool on the right toolbar. Set its exact size with `E`. The outline must be a single closed shape.
4. **Add the electrodes and contact pads.** Use either route from Section 3.5. Contact pads should be large enough for your clip or spring-loaded pin (a few mm square is comfortable).
5. **Route the tracks.** Make `F.Cu` active, press `X`, click on the contact pad, click at each corner, and finish on the electrode. Set the track width from the drop-down list in the top toolbar. Tracks are covered by solder mask automatically. Wider tracks have lower resistance; for impedance work there is no reason to use the minimum width.
6. **Open the mask where metal must be exposed.** Pads made as footprints already have openings. For a larger window (for example over a set of line electrodes), draw a filled rectangle on `F.Mask`.
7. **Add holes and slots.** For bolt or alignment holes, press `A` and place a footprint from the `MountingHole` library. For long holes, see the box below.
8. **Label the board.** Make `F.Silkscreen` active and add text (electrode numbers, cell name, version, date). Future users will thank you.
9. **Check.** Run `Inspect ► Design Rules Checker` and fix any errors. Then look at the board in `View ► 3D Viewer (Alt+3)`: exposed metal appears gold, masked areas in the mask color. This is the most reliable way to confirm that the mask openings are where you intended.

Two ways to make a slot

Oval hole (preferred). In the Footprint Editor, add a pad with type *NPTH*, *Mechanical* and hole shape *Oval*, and set the hole length and width. Manufacturers treat this as a proper slot.

Cut-out on `Edge.Cuts`. Draw a closed shape inside the board outline. It is milled out with the same tool as the board edge, so it cannot be narrower than the tool and its inside corners are rounded.

5 Example 1: Through-Plane Conductivity Cell

Figure 2 shows the layout `TP_cell`. The board carries a single disc electrode connected by one track to a square contact pad at the opposite end; four holes surround the disc. Two such boards are placed face to face with the membrane clamped between the discs (Fig. 3).

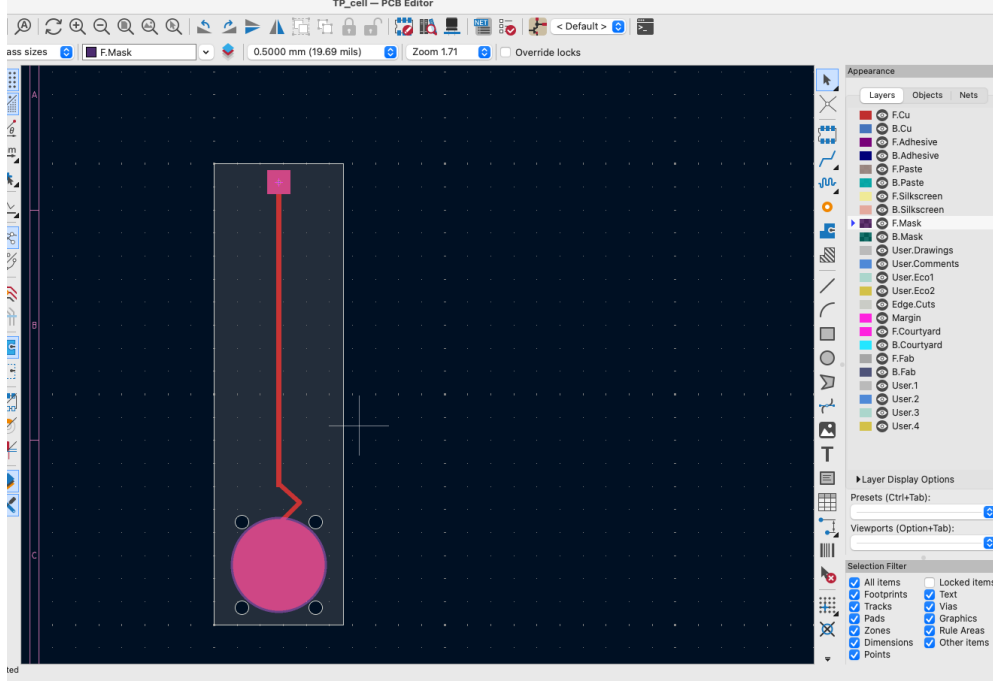


Figure 2: Through-plane cell (`TP_cell`) in the KiCad PCB Editor with `F.Mask` as the active layer. The disc electrode and the square contact pad are exposed (mask openings over copper); the track between them is covered by solder mask.

How it was built in KiCad.

- `Edge.Cuts`: one rectangle, 28×100 mm.
- `F.Cu` + `F.Mask`: a circular pad of diameter 2.0 cm (the electrode) and a square pad of side 5.0 mm (the contact).
- `F.Cu`: one track of width 1.0 mm joining them, with a 45° bend to enter the disc.
- Four holes of diameter 2.8 mm around the disc, used for alignment pins.

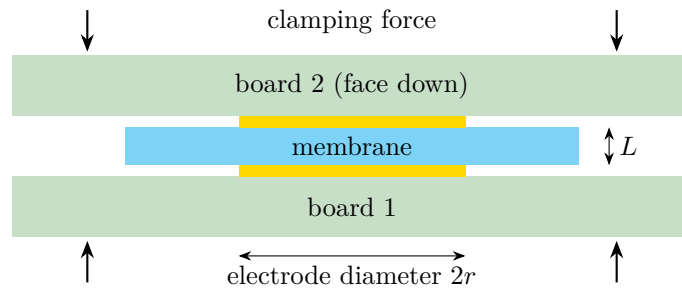


Figure 3: Cross-section of the assembled through-plane cell (not to scale).

6 Example 2: In-Plane Conductivity Cell

Figure 4 shows the layout `IP_cell_long-holes`. Five contact pads along the top edge are connected by five tracks to parallel line electrodes at the bottom of the board. The line electrodes sit inside a rectangular solder-mask window, and four long slots, labeled 1–4 on the silkscreen, are cut through the board between them. The membrane is laid across the electrodes (Fig. 5).

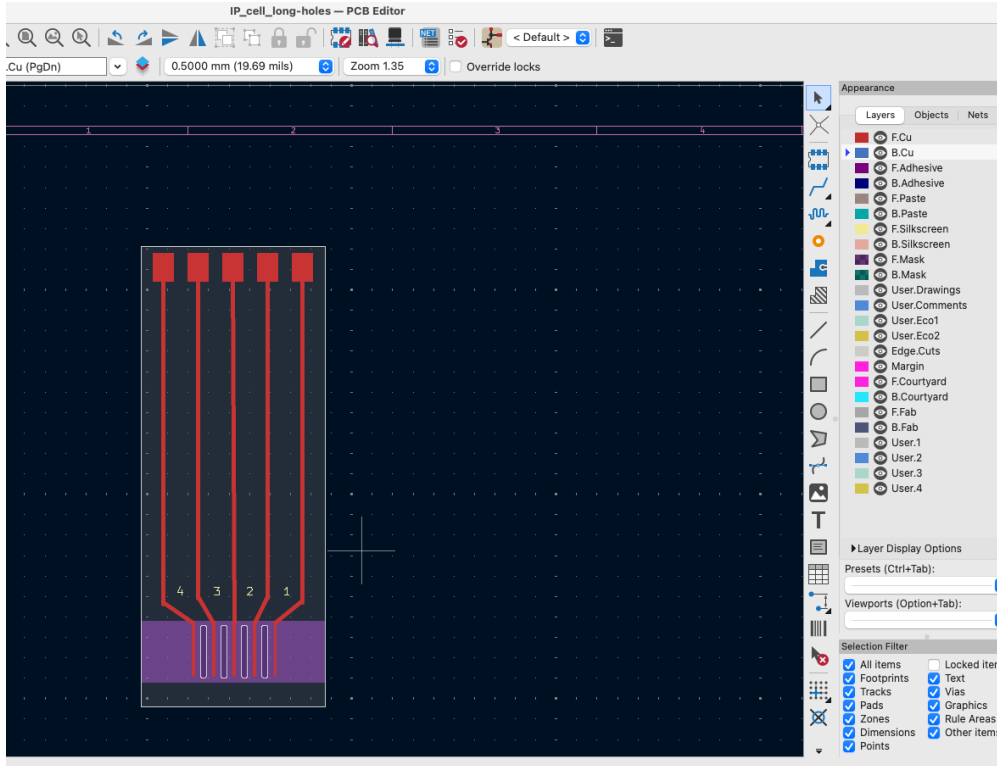


Figure 4: In-plane cell (`IP_cell_long-holes`) in the KiCad PCB Editor. The shaded rectangle at the bottom is the mask opening that exposes the line electrodes; the rounded outlines between the electrodes are slots through the board.

How it was built in KiCad.

- `Edge.Cuts`: one rectangle, 45 mm $\times$ 112.5 mm.
- `F.Cu` + `F.Mask`: five rectangular contact pads, 5.0 mm wide and 7.0 mm high.
- `F.Cu`: five tracks of width 1.0 mm, running straight down and then converging at 45° to a center-to-center spacing of 5.0 mm. The lower ends of the tracks are the electrodes.
- `F.Mask`: one filled rectangle over the electrode region, so the tracks are exposed (and gold-plated) only there.
- Four slots, 1.5 mm $\times$ 13 mm, between adjacent electrodes to let water reach both faces of the membrane between the electrodes.
- `F.Silkscreen`: the numbers 1–4.

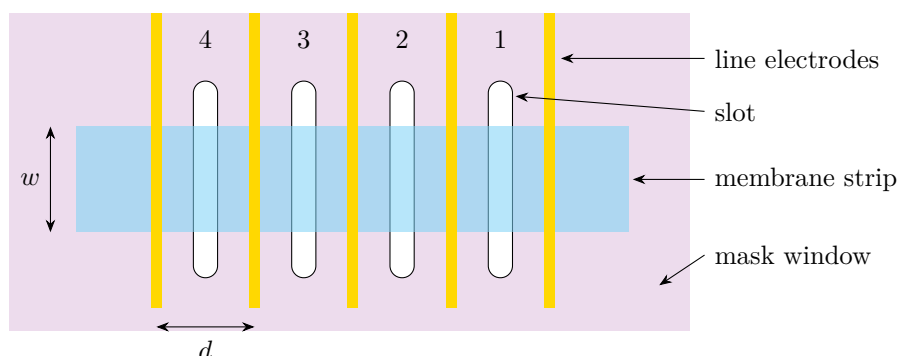


Figure 5: Top view of the electrode region of the in-plane cell with a membrane strip of width w laid across the electrodes (not to scale). d is the distance between two electrodes.

7 Ordering Boards

7.1 What every manufacturer needs

Any board house can make these cells from a set of *Gerber* files (one per layer) plus *drill* files:

1. File ► Fabrication Outputs ► Gerbers (.gbr).
2. Tick the layers F.Cu, B.Cu, F.Mask, B.Mask, F.Silkscreen, B.Silkscreen and Edge.Cuts; choose an output folder; click Plot.
3. In the same dialog click Generate Drill Files and generate them into the same folder.
4. Open the folder in the Gerber Viewer and check each layer, especially the mask openings and the slots.
5. Compress the folder to a .zip file and upload it to the manufacturer.

7.2 Choices that matter for electrochemistry

- **Surface finish: choose ENIG.** The default at many manufacturers is HASL (tin or tin–lead solder), which is unsuitable as an electrode surface. ENIG gives gold over nickel on all exposed copper.
- **Board thickness.** 1.6 mm is standard and rigid enough to clamp.
- **Solder-mask color.** Cosmetic; any color insulates the tracks.
- **Quantity.** Minimum orders are typically three to five boards.

The gold layer is thin

The immersion gold of an ENIG finish is on the order of $0.1\ \mu\text{m}$ thick, over a few micrometers of nickel. It is adequate for contact-type impedance measurements on hydrated membranes, but it is porous and wears with repeated clamping, and the nickel beneath can corrode under sustained polarization or in aggressive electrolytes. Inspect the electrodes periodically and replace boards that look dull or pitted. Some manufacturers offer thicker or hard gold at extra cost.

7.3 Example: OSH Park

The cells in this guide were made by OSH Park (<https://oshpark.com>), which is convenient for small batches. At the time of writing, its two-layer prototype service has the specifications below:

Item	Value
Price	\$5 per square inch, for three copies
Quantity	Multiples of three
Surface finish	ENIG (gold), standard
Solder mask	Purple
Thickness, copper	1.6 mm FR4, 1 oz copper per layer
Minimum track width and spacing	6 mil (0.15 mm)
Minimum drill	10 mil (0.25 mm)
Edge keep-out for copper	15 mil (0.38 mm)
Minimum cut-out	0.068 in (1.7 mm); cut-outs are not plated
Turnaround	Ships in 9–12 calendar days

Uploading. OSH Park accepts either a zip of Gerber and drill files or the KiCad `.kicad_pcb` file itself, which saves the export step. Two caveats apply to direct upload:

- The file is opened with OSH Park’s own installed KiCad version (9.x at the time of writing). A board saved by a newer KiCad may not open; in that case upload Gerbers.
- Silkscreen text may change font unless fonts are embedded in the board file (Board Setup ► Embedded Files ► Embed Fonts).

After upload, the website shows a rendering of every layer. Check the mask openings and slots there before paying: it is the last chance to catch a mistake.

7.4 Other manufacturers

The same Gerber zip works with any other board house (for example JLCPCB, PCBWay or Eurocircuits, or a local supplier). When ordering elsewhere: 1) select ENIG explicitly, since it is usually an option rather than the default; 2) enter that manufacturer’s limits in **Board Setup** and rerun the design-rule check, since minimum track, hole and slot sizes differ; 3) check how slots and cut-outs must be specified, since conventions vary.

8 Further Reading

- KiCad download page: <https://www.kicad.org/download/> and documentation, including “Getting Started in KiCad”: <https://docs.kicad.org>
- OSH Park documentation (KiCad uploads, two-layer service, board outline rules): <https://docs.oshpark.com>
- J. C. Díaz and J. Kamcev, “Ionic conductivity of ion-exchange membranes: Measurement techniques and salt concentration dependence,” *Journal of Membrane Science*, 618, 118718 (2021). doi:10.1016/j.memsci.2020.118718
- J. C. Díaz, D. Kitto and J. Kamcev, “Accurately measuring the ionic conductivity of membranes via the direct contact method,” *Journal of Membrane Science*, 669, 121304 (2023). doi:10.1016/j.memsci.2022.121304